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PII: S1359-8368(96)00028-5

Composites Part B **28B** (1997) 301–308
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1359-8368/97/$17.00

Dynamic pulse buckling of rectangular composite plates

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(Received 25 August 1995; accepted 19 February 1996)

The elastic dynamic buckling of geometrically imperfect rectangular composite plates under a longitudinal compressive pulse is investigated. Specifically, the effects of fiber orientations of angle-ply laminated panels are studied. Geometric nonlinearities due to large deflections, as well as wave propagation effects due to in-plane inertia terms, are included in the analysis. The applied load is either a force or displacement pulse. A numerical solution, through an explicit finite-difference integration scheme, is then developed. Appropriate dynamic buckling criteria are defined for both loading types, and buckling loads are determined for various loading durations and material lay-up configurations. It is found that the dynamic buckling loads are not always higher than the static ones; in some cases there is a range of loading frequencies near the fundamental frequency of the plate where dynamic buckling occurs for lower loads. Buckling under a displacement pulse occurs at a load higher than that for a force pulse of similar duration. Also, the critical axial displacement is not sensitive to the material configuration. Comparisons with results obtained through a finite-element analysis support the conclusions of the present analysis. © 1997 Elsevier Science Ltd

(Keywords: dynamic buckling; angle-ply laminates; rectangular plates)

INTRODUCTION

The dynamic buckling of structures under transient loads has been summarized in two recent monographs by Lindberg and Florence[1] and Simitses.[2] The buckling of rectangular plates under dynamic in-plane compression has been studied for collision impact loading,[3,4] constant rate of loading as in a rigid universal testing machine[5,6] and also for plastic collapse due to high intensity impulsive loads.[1] With a few minor exceptions,[4] it has been observed that the dynamic buckling loads are higher than the static ones. The ratio of dynamic to static buckling loads depends on the duration of the applied pulse and on the initial geometrical imperfection of the plate. Short pulse durations and small imperfections result in higher dynamic buckling loads.

Most of the available studies relate to isotropic plates, very few to orthotropic ones, and, to the best of our knowledge, results of the dynamic pulse buckling of anisotropic plates have been published only recently.[7] The growing use of composite materials in load carrying components of structures which operate in dynamic environments (e.g. gust load on aircraft wings or wave impact on marine vehicle hulls), necessitates research in this area. Hence, the goal of the present study is to investigate the effects of anisotropic material properties on the pulse buckling of imperfect rectangular plates.

ANALYSIS

A rectangular fiber reinforced laminated plate is loaded by a transient in-plane compressive load L_x (see *Figure 1*). Initially, the plate is not perfectly flat, hence, in addition to the in-plane displacements (u, v in the x, y directions), the compressive load generates transient lateral deflection $w(x, y, t)$.

Differential equations

Assuming Kirchhoff thin plate deformation theory and small rotations of the middle surface, and neglecting rotary inertia effects, the equations of dynamic equilibrium are:

$$N_{xx,x} + N_{xy,y} = \rho h \ddot{u} \tag{1}$$

$$N_{yy,y} + N_{xy,x} = \rho h \ddot{v} \tag{2}$$

$$\begin{aligned}&(N_{xx}w_{,x} + N_{xy}w_{,y} - M_{xx,x} - M_{xy,y})_{,x} \\ &\quad + (N_{yy}w_{,y} + N_{xy}w_{,x} - M_{yy,y} - M_{xy,x})_{,y} = \rho h \ddot{w}\end{aligned} \tag{3}$$

where ρ is the average mass density of the plate, h is its uniform thickness, $w_{,x} \equiv \partial w/\partial x, \ddot{u} \equiv \partial^2 u/\partial t^2$ and the force (N) and moment (M) resultants per unit length are:

$$(N_{ij}\,;\,-M_{ij}) = \int_{-h/2}^{h/2} \sigma_{ij}(1; z)\,\mathrm{d}z \tag{4}$$

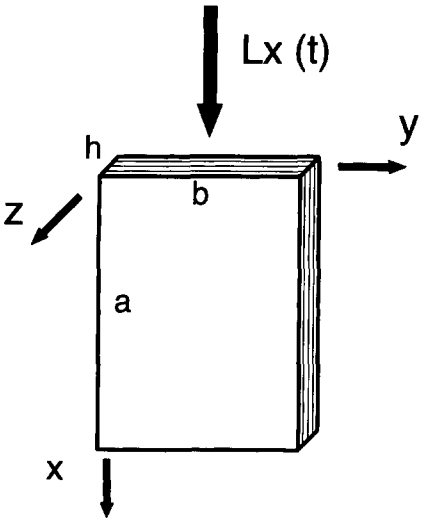

Figure 1 The laminated plate

For a symmetric angle-ply laminate $[n(\pm\theta)]_s$, the load–deformation relations are:

$$\begin{Bmatrix} N_{xx} \\ N_{yy} \\ N_{xy} \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{12} & A_{22} & 0 \\ 0 & 0 & A_{66} \end{bmatrix} \begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ 2\epsilon_{xy} \end{Bmatrix} \tag{5}$$

$$-\begin{Bmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{Bmatrix} = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{Bmatrix} \kappa_{xx} \\ \kappa_{yy} \\ 2\kappa_{xy} \end{Bmatrix} \tag{6}$$

where the strains ϵ_{ij} and curvatures κ_{ij} (for $i;j = x;y$) are:

$$\epsilon_{ij} = \tfrac{1}{2}[u_{i,j} + u_{j,i}) + (w_{,i}w_{,j} - w_{0,i}w_{0,j})] \tag{7}$$

$$\kappa_{ij} = -(w - w_0)_{,ij} \tag{8}$$

and A_{ij} and D_{ij} are the extensional and flexural elastic stiffness of the composite plate. In equation (7), $u_x = u$ and $u_y = v$. Note that the deflection w and the initial geometrical imperfection w_0 are both measured from the plane Oxy, hence the lateral displacement from the undeformed imperfect shape is $w - w_0$.

Through substitution of equations (7)–(8) into (5)–(6) and then to the equations of dynamic equilibrium (1)–(3), three coupled equations of motion are obtained for the transient displacements $u(x,y,t)$, $v(x,y,t)$ and $w(x,y,t)$ of the middle surface of the plate.

Initial and boundary conditions

It is assumed that the plate is initially at rest, hence:

$$u = v = 0; \quad w = w_0; \quad \dot{u} = \dot{v} = \dot{w} = 0, \quad \text{at } t = 0 \tag{9}$$

The shape of the initial imperfection $w_0(x,y)$ must not violate the boundary conditions.

Four boundary conditions must be satisfied along each edge: two for the lateral motion and two for the in-plane constraints. For the purpose of this study, it was decided to adopt one set of constraints, with all the edges straight, laterally clamped and free from in-plane shear. The boundary conditions are then:

$$\begin{aligned} w = w_{,x} = u_{,y} = N_{xy} = 0; \quad &\text{for } x = 0, a \\ u = 0; \quad &\text{for } x = a \end{aligned} \tag{10}$$

$$w = w_{,y} = v = N_{xy} = 0; \quad \text{for } y = 0, b \tag{11}$$

The boundary condition $u_{,y}(x = 0) = 0$ (straight loaded edge) results in a non-uniform distribution of N_{xx}. In addition, there is a loading condition at $x = 0$. If the applied compressive load $L_x(t)$ is a force pulse $F_x(t)$, then:

$$F_x = -\int_0^b N_{xx}|_{x=0}\,\mathrm{d}y \tag{12}$$

but if a displacement pulse $U_x(t)$ is applied, then:

$$u|_{x=0} = U_x(t) \tag{13}$$

Note that the applied compressive force F_x is defined positive in compression.

The initial geometrical imperfection w_0 was assumed as:

$$w_0(x,y) = W_0 \sin^2\left(\frac{\pi x}{a}\right) \sin^2\left(\frac{\pi y}{b}\right) \tag{14}$$

which is compatible with the clamped edge conditions.

The transient load was chosen as a half-sine compressive pulse with a duration T:

$$L_x(t) = \begin{cases} L_m \sin(\pi t/T), & T \geq t \geq 0 \\ 0 & , \quad t \geq T \end{cases} \tag{15}$$

where L_x is either the force (F_x) or the displacement (U_x) pulse at $x = 0$. The chosen pulse allows various frequencies (π/T) and intensities (L_m) of the longitudinal compression.

FDM formulation

The Finite-Difference Method (FDM) was employed to solve the dynamic response of the anisotropic plate. Central difference approximations were used, and the differential equations (1)–(3) resulted in three explicit equations for u, v and w at $t = t + \Delta t$. For numerical stability, the integration time step Δt satisfies:

$$\Delta t \leq \frac{\Delta x}{c_x}; \quad \frac{\Delta y}{c_y} \tag{16}$$

where

$$c_x = \sqrt{\frac{A_{11}}{\rho h}}; \quad c_y = \sqrt{\frac{A_{22}}{\rho h}} \tag{17}$$

are the dilatational wave propagation velocities in the x and y directions, respectively.

RESULTS AND DISCUSSION

The response and dynamic buckling results which are presented here are for typical glass–epoxy symmetric

Table 1 Laminate stiffnesses

	θ [deg]			
	0	15	30	45
A_{11} (MN/m]	151.1	134.2	94.26	53.72
A_{12}	5.78	13.91	30.17	38.29
A_{22}	21.41	21.97	29.44	53.72
A_{66}	7.71	15.84	32.1	40.23
D_{11} [Nm]	201.4	179.	125.7	71.63
D_{12}	7.71	18.55	40.22	51.06
D_{22}	28.55	29.29	39.25	71.63
D_{16}	0.	30.28	42.15	32.41
D_{26}	0.	2.13	13.99	32.41
D_{66}	10.29	21.12	42.8	53.64

angle-ply laminates, with $(\pm\theta)_s$, where $\theta = 0°$, 15°, 30° and 45°. The mass density is $\rho = 2200\,\mathrm{kg/m^3}$ and the total thickness of the four-layer laminate is $h = 4\,\mathrm{mm}$. The elastic stiffness coefficients A_{ij} and D_{ij} are presented in *Table 1*.

The plate dimensions are $a = b = 1\,\mathrm{m}$, hence $a/h = 250$ which satisfies the thin plate assumptions. The amplitude of the initial imperfection is $W_0 = 1\,\mathrm{mm}$, hence $W_0/h = 0.25$.

The pulse duration T is selected relative to $T_b/2$, where T_b is the period of the fundamental flexural natural frequency of the plate. When T is much shorter than $T_b/2$ the load is impulsive, but when T is much longer than $T_b/2$ the load is quasi-static. In the intermediate range, for which $T \approx T_b/2$, the load is referred to as "dynamic". The natural periods of the plates in this study fall within the range $T_b \approx 40$–$60\,\mathrm{ms}$. Hence, the response of each plate was analyzed for pulse durations from $T = 5$ to $150\,\mathrm{ms}$, which cover a wide range of relative frequencies, from impulsive to long duration ("quasi-static") loads. *Figures 2–4* present deflection histories at the center of the $(\pm 45°)_s$ plate under a force pulse, for impulsive (*Figure 2*), dynamic (*Figure 3*) and quasi-static compression (*Figure 4*). The response reaches its peak at different times. For low intensity pulses (much below buckling level), the peak deflection under impulsive load is obtained after the pulse was released; under quasi-static compression the maximum occurs approximately when the load reaches its peak; whereas under dynamic pulse the peak deflection occurs shortly before the load release. For high intensity pulses (near buckling level), the peak deflection is reached earlier and, as will be discussed later, the response is different. Similar trends were observed also for other ply angles as well as for plates under displacement pulse.

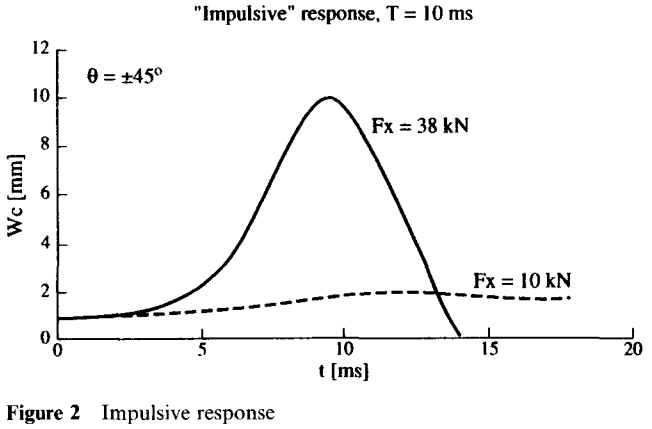

Figure 2 Impulsive response

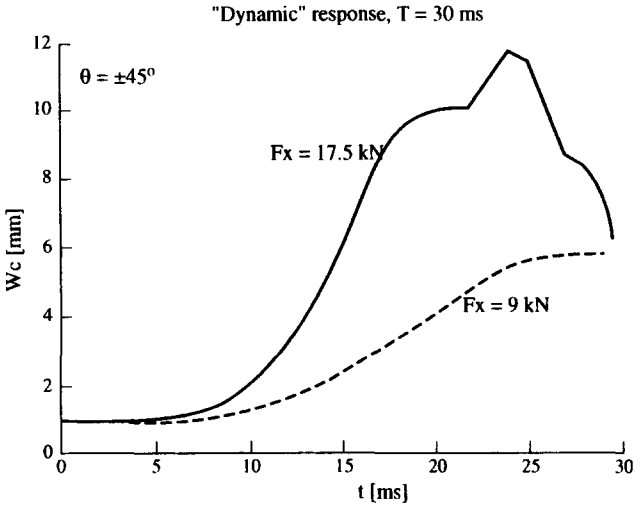

Figure 3 "Dynamic" response

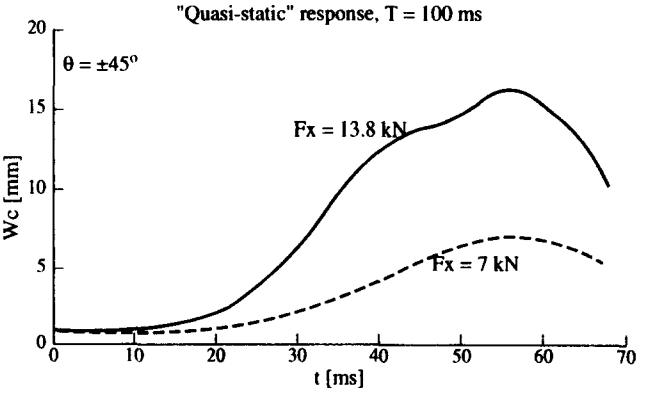

Figure 4 Quasi-static response

Buckling criteria

Bifurcation buckling does not occur in dynamic pulse buckling of plates. Rather, its behavior involves deflection growth of an initially imperfect plate. Buckling loads, namely the pulse intensities that cause dynamic buckling, may be extremely high for nearly perfect plates but much lower for plates with appreciable imperfection. It is, therefore, necessary to define dynamic buckling and establish criteria to determine dynamic buckling loads. Note that the term "dynamic buckling" refers here only to pulse buckling and does not relate to parametric excitation instabilities.

A widely accepted definition of dynamic buckling of imperfect structures states that buckling occurs when a small increase in the load intensity results in an unbounded growth of the deflections.[8] This definition

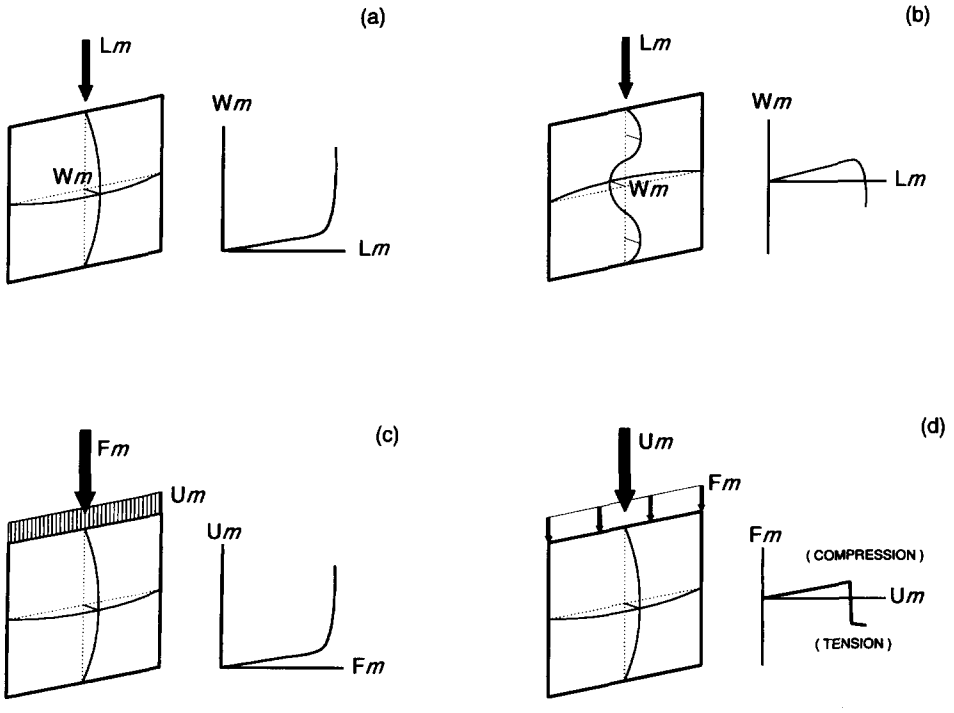

Figure 5 Buckling criteria

is not generally applicable to every structure, but it may be adapted to suit various problems.

Four different buckling criteria were considered in this study. The first criterion (*Figure 5(a)*) relates the peak lateral deflection w_m to the pulse intensity L_m. Buckling occurs when, for a given pulse shape and duration, a small increase in the pulse intensity causes a sharp increase in the rate of growth of the peak lateral deflection. The advantages of this criterion are that it is applicable to both displacement and force loading types and it may be used for a wide range of pulse frequencies. However, for very short pulse durations, where high load intensities are required for buckling, the deflection shapes change to short wavelength patterns which are associated with smaller peak deflections and this criterion is then misleading. Hence, the second criterion (*Figure 5(b)*) associates dynamic buckling with a pattern of short wavelength deflection shape. In this case, buckling occurs when a small increase in the pulse intensity causes a decrease in the peak lateral deflection. This criterion is relevant to impulsive loads only and may be used to complement the first criterion. The third and fourth criteria are both collapse-type buckling criteria, relating the intensity of the applied load to the peak response at the loaded edge $x = 0$. The third buckling criterion applies to a force pulse (*Figure 5(c)*). Buckling occurs when a small increase in the force intensity F_m causes a sharp increase in the peak longitudinal displacement U_m at $x = 0$. It occurs because the structural resistance to the inplane compression diminishes when the dynamic lateral deflections grow rapidly. The fourth buckling criterion applies to a displacement pulse (*Figure 5(d)*). Buckling occurs when a small increase in the pulse displacement intensity U_m

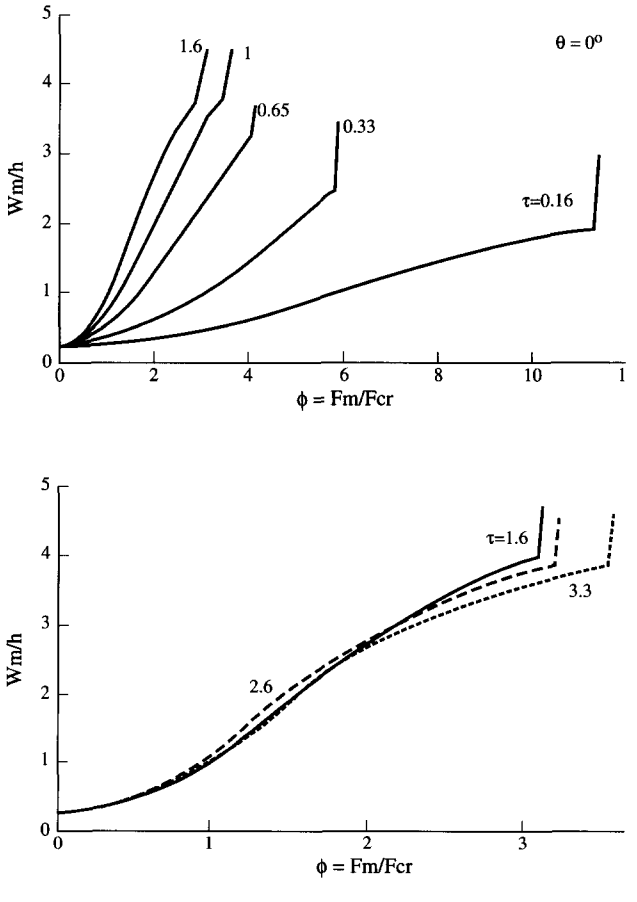

Figure 6 (a) Peak deflection *versus* load intensity for various pulse durations. (b) Same for slower loads

causes a transition of the peak reaction force F_m at $x = 0$ from compression to tension. The transition occurs when the tensile force required to hold the deforming plate at the prescribed U_m is larger than the peak compression at the loaded edge.

FDM results

For each pulse duration, results were obtained for increasing intensities of the compressive load until plate buckling was obtained. *Figure 6* presents an example (for $\theta = 0$) of the changes in the maximum deflection w_m of the plate that occur when the intensity of the force pulse F_m is increased. The results are presented in a nondimensional form, using

$$\phi = \frac{F_m}{F_{cr}}; \quad \tau = \frac{T}{T_b/2} \tag{18}$$

where F_{cr} is the bifurcation static buckling load of the perfectly flat plate,[9] ϕ is the dynamic load ratio and τ is the nondimensional pulse duration. At low load levels the plate stiffness decreases and the rate of growth of the deflection increases. For higher loads, due to in-plane stretching, there is a nonlinear stiffening, and finally the response diverges very rapidly. The load that causes the sudden increase of the lateral deflections is defined as the buckling load (first buckling criterion). Note that this buckling occurs after the nonlinear stretching and it is, therefore, not comparable with the linear static buckling load. For long duration loads, as in *Figure 6(b)* for $\tau \geq 1.6$, the initial stiffness reduction occurs for $\phi < 1$, while the ultimate nonlinear dynamic buckling occurs for $\phi > 1$. The buckling loads for $\tau \approx 1.6$ and 2.6 are lower than that for the quasi-static $\tau \approx 3.3$, which indicates that dynamic buckling loads lower than the static one are possible.

A similar behavior was observed for plates under displacement pulse loads (*Figure 7*). However, the resulting axial force at buckling (*Figure 8*) is appreciably higher than the buckling load under force pulse (*Figure 6(a)*). This difference is due to the stabilizing effect of the displacement pulse, which holds the loaded edge at the prescribed displacement and forces it back to zero after the removal of the pulse. On the other hand, under force pulse, unrestrained displacements and eventual collapse are allowed.

Obviously, the buckling force depends very much on the frequency of the pulse. Very large intensities are required to cause dynamic buckling in the impulsive

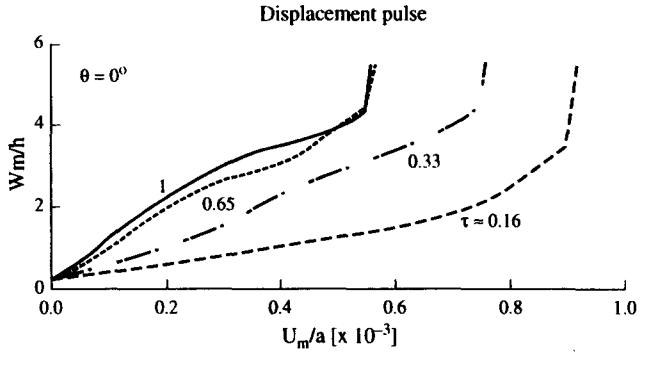

Figure 7 Peak deflection under displacement load U_m for various pulse durations

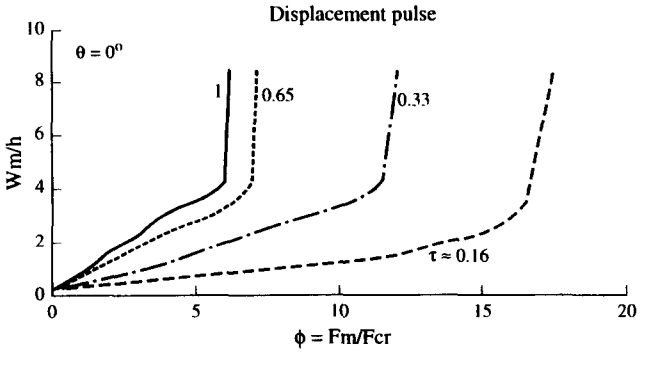

Figure 8 Peak deflection and reaction force at displaced edge

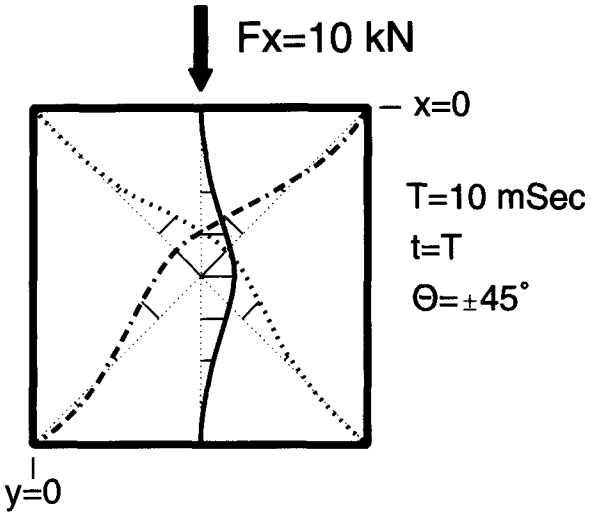

Figure 9 Deflection shape under low load intensity

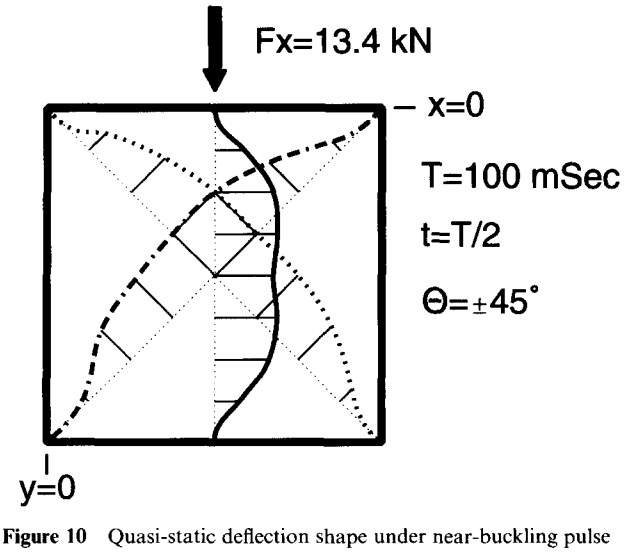

Figure 10 Quasi-static deflection shape under near-buckling pulse

range ($\phi > 10$ for $\tau = 0.16$). In the dynamic range the buckling force drops rapidly to a level near the quasi-static buckling. Moreover, as presented in *Figure 6(b)*, it may even be lower than the static value. This is apparently due to the dynamic amplification which occurs when the pulse frequency is near the resonance frequency of the plate.

Before the ultimate divergence of the deflection occurs, the stiffness appears to become very large, especially for the shorter pulse durations. There are two possible reasons for the increased stiffness. The first reason is the inplane tension and the associated nonlinear stiffening that were mentioned before. The second possible reason is a change in the mode of lateral deflection. This change is illustrated in *Figures 9–11*, where deflection shapes are presented along the two diagonals and center line. When low intensity loads are applied (*Figure 9*) the maximum deflection shape is actually a simple magnification of the initial imperfection, even when the pulse frequency is very high (impulsive). For high intensity loads, however, where the compression pulse is near the buckling level, there is a change in the deflection pattern, which is very mild in the quasi-static range (*Figure 10*), but very significant in the dynamic and impulsive (*Figure 11*) realms. The initial half-wave shape collapses, and a pattern of three half-waves is generated. Similar changes in deflection patterns were observed also for the other lay-up configurations and under both force and displacement pulse loads. Note that because of the different coupling stiffnesses D_{16} and D_{26}, the deflections along both diagonals of the $\theta = \pm 45°$ plate, as well as other $\theta \neq 0°$ plates, are different.

Summaries of the dynamic buckling data for all the plates are presented in *Figures 12* and *13* for force and displacement pulses, respectively. They show that the dynamic buckling loads are near the static ones for the dynamic and quasi-static pulse frequencies, but significantly higher in the impulsive range. The plots of *Figure 12(a)* are normalized according to the definitions of equation (18), using the overestimated orthotropic F_{cr}

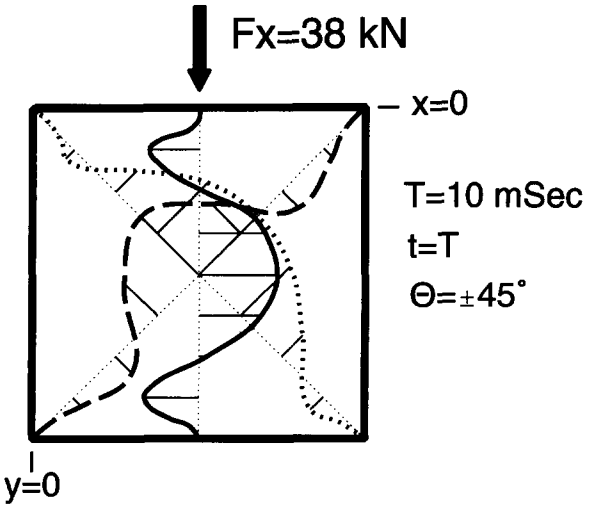

Figure 11 Impulsive deflection shape under near-buckling pulse

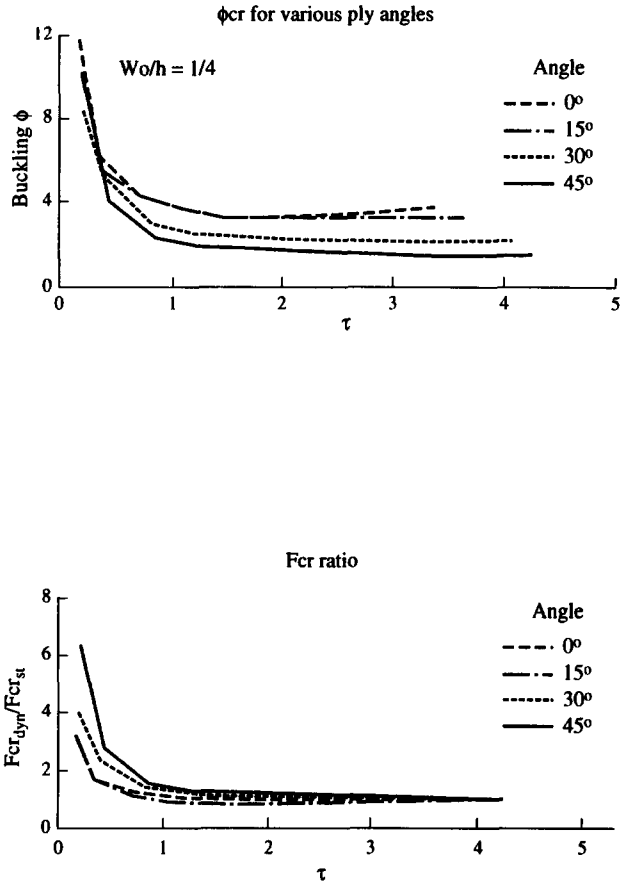

Figure 12 (a) Dynamic buckling load *versus* normalized pulse duration for various laminate lay-ups under force pulse. (b) Dynamic to quasi-static nonlinear buckling load ratio *versus* normalized pulse duration for various laminates

for ϕ^9, so only the plot for $\theta = 0°$ is "exact". However, when the dynamic buckling load is normalized relative to the corresponding nonlinear quasi-static buckling load, the true ratio of dynamic to static buckling loads, as presented in *Figure 12(b)*, is obtained. The plots of *Figure 12(b)* show that, in the impulsive range, the ratio is highest for the $(\pm 45°)_s$ lay-up and lowest for $\theta = 0°$ and 15°. Similar trends are presented in *Figure 13* for plates under displacement pulse.

The above results were based on the first buckling criterion. Buckling loads obtained through the collapse-

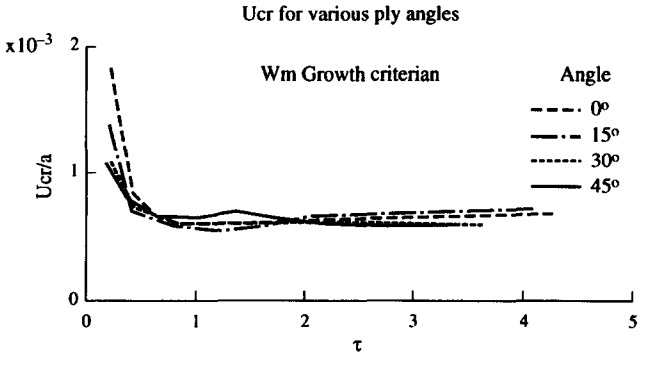

Figure 13 Dynamic buckling load *versus* pulse duration for various laminate lay-ups under displacement load

type buckling criteria were not much different since the sharp increase in the peak displacement $u_m|_{x=0}$ is associated with that for w_m. However, when the second buckling criterion, which is associated with a transition to a short wavelength deflection pattern, is used, the buckling loads in the dynamic and impulsive ranges are lower (compare *Figure 14* to *Figure 13*). For conservative buckling results it may, therefore, be advisable to employ this criterion.

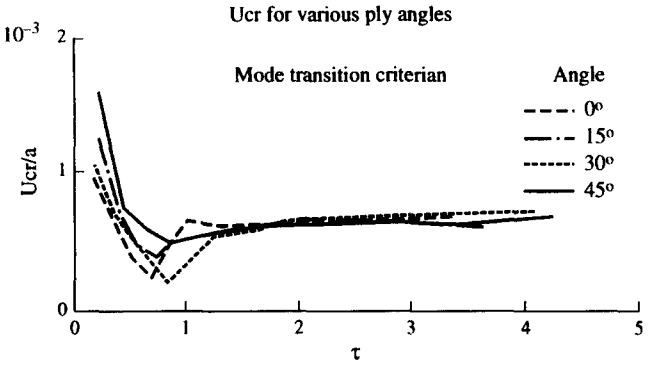

Figure 14 Dynamic buckling load *versus* pulse duration for various laminate lay-ups under displacement load, using the mode transition buckling criterion

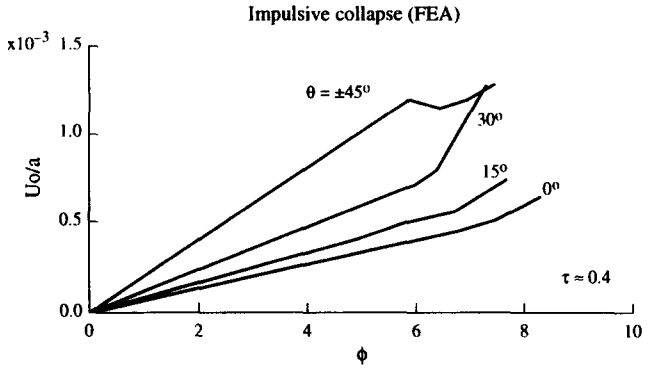

Figure 15 Finite element analysis results of loaded edge displacement *versus* impulsive force intensity for various ply angles

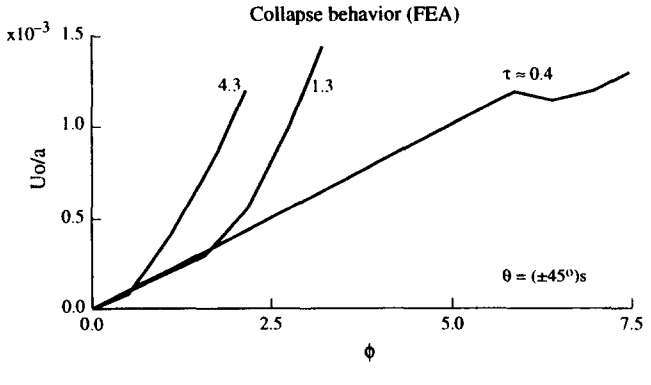

Figure 16 Finite element analysis results of loaded edge displacement *versus* force intensity for a range of pulse durations

FEA results

A commercial finite element analysis code (ANSYS®) was used to compare the results of this study for force-controlled pulse loads. The orthotropic layer properties of the four-layer symmetric angle-ply laminated plates are: $E_L = 37.4$ GPa, $E_T = 5.1$ GPa, $\upsilon_{LT} = 0.28$ and $G_{LT} = 1.9$ GPa. Some results for various lay-up configurations and pulse durations are presented in *Figures 15–16*. The dynamic buckling loads are within 15% of the FDM results (see *Figure 17*). Also, the impulsive short wave-length deformation pattern (*Figure 18*) exhibits the same features of three half-waves as obtained by the FDM results (*Figure 11*).

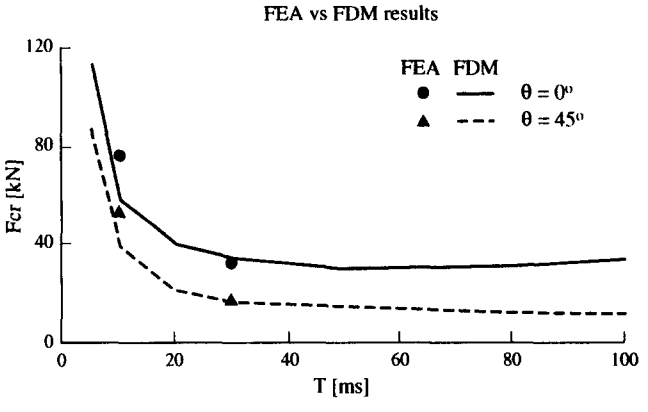

Figure 17 Comparison of finite element analysis dynamic buckling results with present finite difference method results for two extreme ply angles

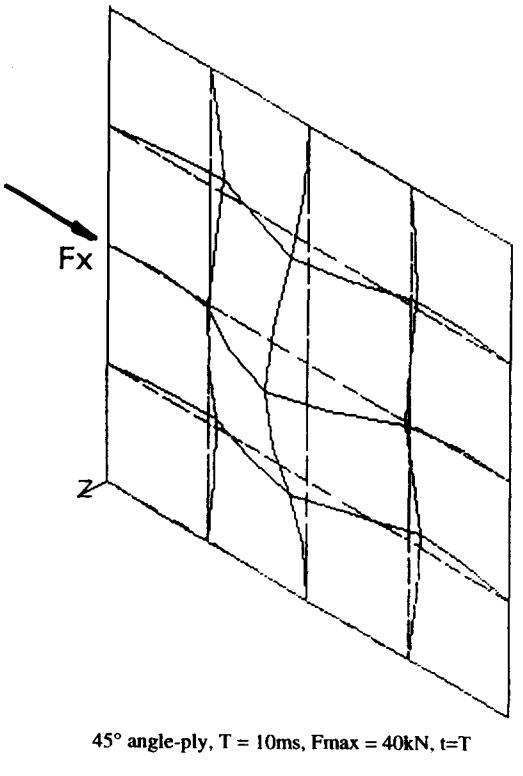

Figure 18 Finite element analysis impulsive ($\tau = 0.4$) deflection shape ($\phi = 4.3$) of $(\pm 45°)_s$ laminate

CONCLUSIONS

The dynamic buckling of geometrically imperfect angle-ply composite plates under longitudinal compressive pulse was investigated. The loads are either force-controlled or displacement-controlled. It was shown that laminate configurations which provide higher static buckling resistance are also stronger in pulse buckling, but the ratio depends on the laminate lay-up. The dynamic buckling loads are not always higher than the static ones; there is a range of pulse frequencies near the fundamental frequency of the plate where dynamic buckling may occur at slightly lower loads. In general, pulse buckling loads in the dynamic realm are very close to the static ones; and only under impulsive loads higher intensities are needed to cause buckling. Under a displacement pulse buckling occurs at a load higher than that for a force pulse with similar duration. Comparisons with FEA results support the conclusions of the present analysis. One should be reminded that the basic assumption here is that the material is linear elastic. Hence, the results are for overall buckling only. Practically it is possible, for example, that prior to buckling the material will disintegrate under the high intensity loads of the impulsive range. Also, local fiber buckling may change the compressive stiffness properties of the laminate and, therefore, the overall buckling behavior. Further studies for effects of transverse shear and rotary inertia, as well as different boundary conditions and initial imperfections, are needed.

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